Machine Intelligence Research
www.mi-research.net

22(2), April 2025, 219-238
DOI: 10.1007/s11633-024-1492-6

Attention Detection Using EEG Signals and Machine Learning: A Review

Qianru Sun[1,2] Yueying Zhou[1,2] Peiliang Gong[1,2] Daoqiang Zhang[1,2]

[1] Key Laboratory of Brain-Machine Intelligence Technology, Ministry of Education, Nanjing University of Aeronautics and Astronautics, Nanjing 211106, China

[2] College of Computer Science and Technology, Nanjing University of Aeronautics and Astronautics, Nanjing 211106, China

Abstract: Attention detection using electroencephalogram (EEG) signals has become a popular topic. However, there seems to be a notable gap in the literature regarding comprehensive and systematic reviews of machine learning methods for attention detection using EEG signals. Therefore, this survey outlines recent advances in EEG-based attention detection within the past five years, with a primary focus on auditory attention detection (AAD) and attention level classification. First, we provide a brief overview of commonly used paradigms, preprocessing techniques, and artifact-handling methods, as well as listing accessible datasets used in these studies. Next, we summarize the machine learning methods for classification in this field and divide them into two categories: traditional machine learning methods and deep learning methods. We also analyse the most frequently used methods and discuss the factors influencing each technique's performance and applicability. Finally, we discuss the existing challenges and future trends in this field.

Keywords: Attention detection, electroencephalogram (EEG), machine learning, deep learning, brain-computer interface.

Citation: Q. Sun, Y. Zhou, P. Gong, D. Zhang. Attention detection using EEG signals and machine learning: A review. *Machine Intelligence Research*, vol.22, no.2, pp.219–238, 2025. http://doi.org/10.1007/s11633-024-1492-6

1 Introduction

Attention refers to the capacity to select, modulate, and sustain a focus on the most relevant information for behavior, and it plays an essential role in all perceptual and cognitive operations[1]. Attention helps individuals allocate cognitive resources to specific tasks and ignore background noise. Furthermore, attention serves as an important reference for the investigation of neuroscience and many other disciplines[2]. Therefore, attention has become a popular research topic in both academia and industry.

Despite its importance, attention has always been an ambiguous concept, making it hard to give a precise definition. According to Knudsen's attentional model, "attention is considered as the result of three coordinated processes: (A) monitoring and evaluation of ongoing cognitive processes, (B) excitation of task-relevant processes, and (C) inhibition of task-irrelevant processes[3]". Attention can be divided into several submodalities, including selective attention, spatial attention, visual attention, sustained attention, and vigilance[3].

Due to the importance of attention in daily life, the analysis of attention has many practical applications. In response to this need, researchers have developed several techniques to measure attention state, including eye blinking, functional magnetic resonance imaging (fMRI), muscle movement, and electroencephalogram (EEG). Among these techniques, EEG is a noninvasive, effective, and convenient method for analysing attention. EEG records the electrical signals produced by the brain, reflecting changes in the brain activity over time. By analysing EEG signals, researchers can discriminate between attentive states, monitor variations in visual sustained attention, and decode auditory and spatial attention. Thus, applying EEG signals in attention detection and analysis has a lot of practical significance in attention studies.

There is a long history of research on attention using EEG signals. In the 1980s, several studies found evidence that the gamma band may be related to the attention or intention of movement[4]. Since then, researchers have discovered associations between attention and EEG signals, demonstrating that visual attention, spatial attention, auditory attention, and attention level can all be measured through the analysis of EEG signals. Many researchers have applied this technique to practical applications. For example, Zhang and Etemad[5] estimated the vigilance and attention level of drivers to ensure transporta-

Review
Manuscript received on September 11, 2023; accepted on January 8, 2024; published online on January 7, 2025
Recommended by Associate Editor Junzhou Huang
Colored figures are available in the online version at https://link.springer.com/journal/11633

tion safety. Narayanan et al.[6] and Su et al.[7] have used EEG sensors to detect auditory attention and applied this information to neuro-steered hearing devices. Reichert et al.[8] used overt and covert attention to provide reliable control of the brain-computer interface (BCI) systems. Miranda et al.[9] found that the detection of attentive states can aid in the diagnosis of common diseases such as conduct disorder or attention deficit hyperactivity disorder (ADHD). There is also a trend toward portable and mobile EEG devices. For instance, O′Sullivan et al.[10] demonstrated that it is possible to interpret a subject′s auditory attention by analysing EEG data from a single trial. Jeong et al.[11] designed an attention state classification system using an in-ear EEG device.

In recent years, many authors have conducted reviews of EEG analysis from various perspectives, covering areas such as emotion recognition[12], disease diagnosis[13], and motor imagery[14]. However, relatively few surveys on the use of EEG signals in attention detection have been conducted. Some surveys have focused on specific aspects of attention. Belo et al.[15] surveyed auditory attention and its future applications. Alickovic et al.[16] reviewed auditory attention detection methods. Other studies have mentioned attention detection under specific conditions. Souza and Naves[2] surveyed attention detection in a virtual environment. Vasiljevic and De Miranda[17] reviewed attention detection methods in BCI games. However, there does not exist a comprehensive and systematic review that covers in-depth the application of machine learning algorithms for EEG-based attention detection.

To fill this gap, this survey offers an up-to-date, comprehensive, and insightful overview of attention detection using EEG signals. Rather than focusing on specific aspects of attention, we provide a broad overview of this field. To achieve this goal, we summarize the methods used at different stages of attention detection, discuss the existing problems and future trends in this field, and provide a comprehensive summary of research suggestions for future researchers. Fig. 1 is an overview of attention detection using EEG signals, which includes six steps: an experimental paradigm design, in which the task and the stimuli are defined and presented to the subjects; EEG data acquisition, where EEG data are recorded from the subjects using electrodes; preprocessing, in which artifacts such as eye movements and muscle activity are removed and the EEG signals are filtered and segmented; feature extraction, in which relevant features such as power spectrum, entropy, and coherence are extracted from the EEG signals; classification, in which machine learning algorithms are applied to classify the attention levels based on the features; feedback, in which the classification results are presented to the users. Our contributions can be summarized as follows:

1) We systematically and comprehensively reviewed papers related to attention detection using EEG signals, summarizing each step.

2) We reproduce deep learning and machine learning methods and implement them in experiments conducted on three public datasets to compare the characteristics of different methods.

3) We propose best practice recommendations and discuss existing problems and future trends.

This paper is organized as follows: Section 2 introduces the fundamental background information about attention categories. Section 3 covers the data sources, common paradigms, EEG recording steps, preprocessing methods, and mainstream classification methods. We keep a primary focus on the utilization of machine learning methods in EEG-based attention detection. Section 4 offers best practice recommendations, discusses current problems in this area, and suggests possible directions for future research. Finally, Section 5 provides the conclusion of the paper.

2 Methods

2.1 Paper selection strategy

We collected papers from platforms such as Google

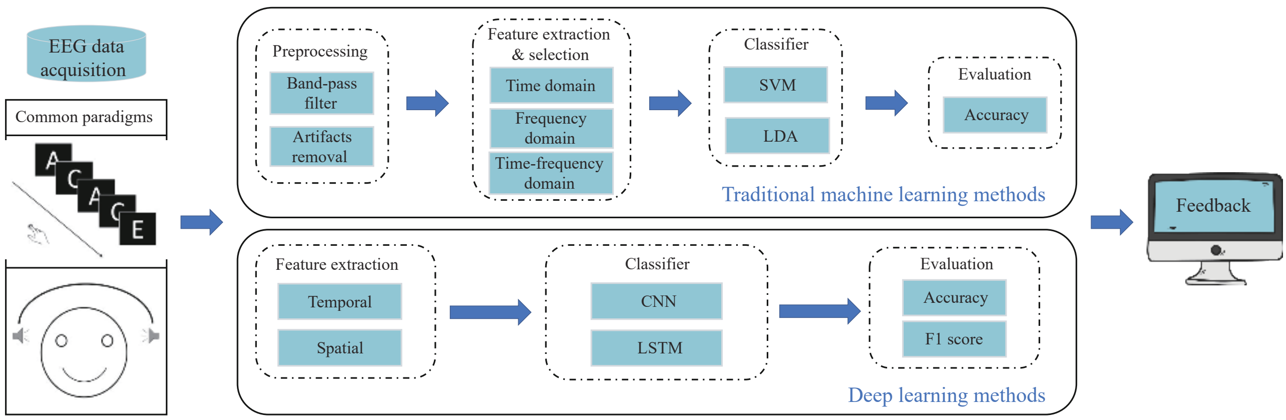


Fig. 1 Flowchart of attention detection using EEG signals

Scholar, PubMed, and Web of Science. During the collection process, we used various combinations of keywords including: ("Attention" OR "Attention Detection" OR "Distraction" OR "Attention Level" OR "Auditory Attention") AND ("EEG" OR "Electroencephalography") AND ("BCI" OR "Brain-Computer Interface"). We removed any duplicate papers that appeared in multiple selections. Our paper collection was guided by the following criteria:

1) The paper must focus on the study of attention and use EEG as the primary technique.

2) Due to our limited research time and to ensure timeliness, we reviewed only studies published between January 1, 2017, and March 1, 2023. Studies published before 2017 were only briefly reviewed in the introduction.

3) To align with current scientific trends, inclusion was limited to papers published in authoritative journals with an impact factor greater than 4.0.

To meet these criteria, we first limited our search to papers published within a specified time frame, resulting in 20 186 papers, after a rapid review of titles and abstracts, 187 papers meeting the criteria were downloaded. We selected 100 articles from premier research journals that met criterion 3) and excluded those with little relevance to attention, leaving us with 86 papers. These 86 papers formed the basis of our research. Notably, we selected 42 papers that focused specifically on EEG analysis methods and examined in detail how they analysed EEG signals.

2.2 Attention category

Due to the complexity of defining attention, studies on the subject often focus on specific aspects of attention mechanisms rather than attempting to encompass all of its facets. We have compiled research questions about attention from the past few years and have created a visual representation of our findings in Fig. 2.

As shown in Fig. 2, the majority of papers on attention published in the last five years have focused primarily on attention level classification (30%) and auditory attention (25%). There have also been many studies on attention itself, including research on attention allocation, attention orienting, and attention mechanisms. The remaining studies have investigated spatial attention (9%), visual attention (7%), sustained attention (5%), vigilance (5%), and selective attention (1%).

Research on attention level classification mainly involves distinguishing between different levels of attention, including distraction detection[18], mind wandering detection[19], attention state classification[20], and attention enhancement[21]. Detecting attentive states can aid in the diagnosis of common diseases such as ADHD and conduct disorder[13, 22]. Auditory attention is particularly important for hearing aid devices[23]. Vigilance is essential

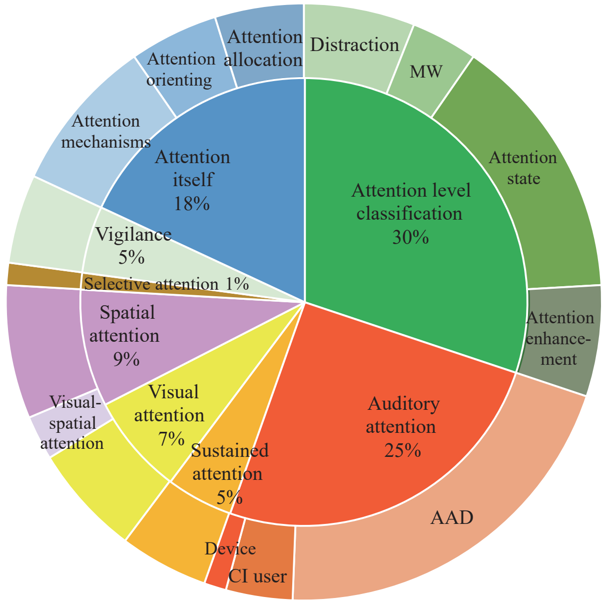


Fig. 2 This figure provides an overview of different fields of attention and their percentages in the last five years. Here MW denotes Mind Wandering, AAD denotes Auditory Attention Detection, and CI user denotes Cochlear Implant user.

for ensuring transportation safety[24]. Both overt and covert attention can provide reliable control of the BCI systems[8]. Therefore, attention detection has many practical uses in daily life.

3 Result

This article focuses on attention detection based on EEG signals and the algorithms involved. We begin by reviewing the common datasets and paradigms used in this field, as well as the EEG devices and electrode configurations employed for recording. Then, we describe the preprocessing, feature extraction, and classification methods applied to the EEG data.

3.1 Common datasets

One of the challenges in EEG analysis is obtaining high-quality and reliable data[25]. There are two ways to obtain data: using public datasets or collecting data by oneself. In this section, we introduce some of the public datasets that are available for EEG attention research. According to our statistics, approximately 28.3% of the articles chose to use public datasets, while the remaining 71.7% of articles chose to collect EEG data by themselves, and 10.9% of articles collected data and chose to make them public.

As shown in Table 1, most of the datasets, such as the KUL[27] dataset and the DTU[28] dataset, focus on auditory attention detection. These two datasets are the most well-known and widely used in many studies. The KUL dataset is recorded by KULeuven and contains both raw and pre-processed data. The DTU dataset was proposed

Table 1 Publicly available datasets

Name/source	NoS	Recording device	Sampling rate	NoE	Usage	Electrodes
SEED-VIG[26]	23	The Neuroscan system	1 000 Hz	17	Vigilance estimation in the driving scene	FT7, FT8, T7, T8, TP7, TP8, CP1, CP2, P1, PZ, P2, PO3, POZ, PO4, O1, OZ, O2
KUL[27]	16	BioSemi ActiveTwo system	8 192 Hz	64	AAD	64 electrodes
DTU[28]	18	BioSemi ActiveTwo system	512 Hz	64	AAD	64 electrodes
Reichert et al.[29]	18	BrainAmp DC Amplifier (Brain Products GmbH, Germany)	250 Hz	14	Spatial attention + BCI control	O9, O10, CP1, CP2, Pz, P3, P4, P7, P8, PO3, PO4, PO7, PO8, Oz
Zhang et al.[30]	30	BrainAmpEEGamplifier	1 000 Hz	28	Attentive state	Fp1, Fp2, Afz, AFF5h, AFF6h, F1, F2, FC5, FC1, FC2, FC6, T7, C3, Cz, C4, T8, CP5, CP1, CP2, CP6, TP9, P7, P3, Pz, P4, P8, TP10, POz, O1, O2
Acı et al.[20]	6	The Epoc headset	128 Hz	7	Attention states classification	F3, Fz, F4, C3, Cz, C4, T3, T4, T5, T6, Pz
Dhindsa et al.[31]	16	The Large Interactive Virtual Environment (LIVE) Lab	–	16	Mind wandering	Fp1, Fpz, Fp2, F7, F3, F4, F8, T7, C3, C4, T8, P7, P8, O1, Oz, O2
Jaeger et al.[32]	21	Easycap, Herrsching, Germany	500 Hz	94	AAD	94 electrodes
Geravanchizadeh and Gavgani[33]	40	A BioSemi ActiveTwo system	512 Hz	128	AAD	128 electrodes
Ciccarelli et al.[34]	11	A Neuroscan 64-channel	1 000 Hz	64	AAD wet electrodes	64 electrodes
Ciccarelli et al.[34]	11	A Wearable Sensing DSI-24 system	300 Hz	18	AAD dry elctrodes	18 electrodes
Das et al.[35]	28	The BioSemi ActiveTwo system	8 192 Hz	64	AAD	64 electrodes

Note: NoS denotes the number of subjects; NoE denotes the number of electrodes used; AAD denotes auditory attention detection; A dashed line in the table signifies that the corresponding aspect was not mentioned in the respective paper.

by Fuglsang et al.[36] in their work. The remaining datasets cover other problems of the attention field, such as vigilance estimation, spatial attention classification, attention state, distraction, and mind wandering. Most of the datasets record all the electrodes for the convenience of usage, and the number of subjects ranged from 6 to 40, with an average of 15.

3.2 Common paradigms

This section introduces the common experimental paradigms for AAD problems and attention level classification problems and provides an analysis of the number of subjects. Fig. 3 presents a schematic diagram that illustrates two common experimental paradigms, one for AAD and the other for attention-level classification.

1) Auditory attention detection

Most researchers use dichotic listening tasks to study auditory attention[37]. In dichotic listening tasks, two different auditory stimuli are presented to participants at the same time. Participants are required to ignore stimuli from one ear while focusing on the stimuli from the other ear during the experiment. The stimuli are usually speech, consisting of several native speakers talking simultaneously in an anechoic environment. The speech content could be Dutch stories (KUL dataset) or other meaningful words. These speech streams are usually nor-

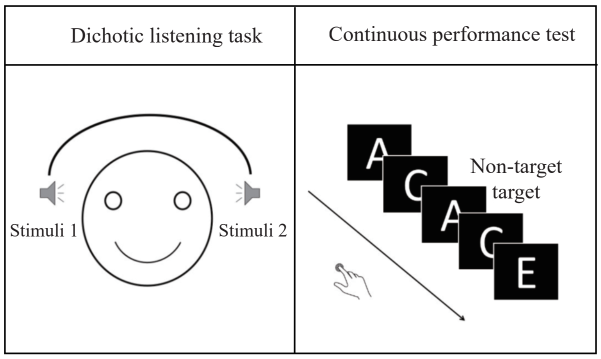


Fig. 3 Two common paradigms: the dichotic listening task for the AAD problem and the continuous performance test for the attention level classification problem.

malized to the same RMS value.

2) Attention level classification

Continuous performance test (CPT) is a popular method for measuring attention[38]. CPT was first developed by Rosvold et al.[39] in 1956. Participants are required to respond differently to make different responses to different stimuli, and a high level of concentration is needed to finish this task. There are several variations of the CPT, including the X-CPT, AX-CPT, and SART[40]. Among frequent non-target stimuli, the X-CPT requires individuals to respond to rare target stimuli. The AX-

CPT adds a cue stimulus before the target or non-target stimulus. The SART requires subjects to respond to frequent target stimuli and inhibit responses to rare non-target stimuli.

The discrimination/selection response (DSR) task is also used in the study of attention[41]. The DSR task consists of 3 sessions, each with 3 series of 20 trials. Each series includes an instruction period, a task period, and a rest period. During the task period, “O” or “X” symbols are shown randomly, with “O” appearing 30% of the time and “X” appearing 70% of the time. Participants need to press the corresponding buttons when the symbol is shown on the screen. During the rest period, subjects are required to relax and avoid excessive eye movements. The task period is used as an attentive state, while the rest period is used as a non-attentive state.

3) Subject

The number of participants varies greatly among the papers, ranging from 5 to 331. Fig. 4 provides detailed information about the number of subjects in each paper. The average number of participants is 25, while the median number of participants is 16. 25.7% of studies use fewer than 10 participants, which may limit the validity of their results.

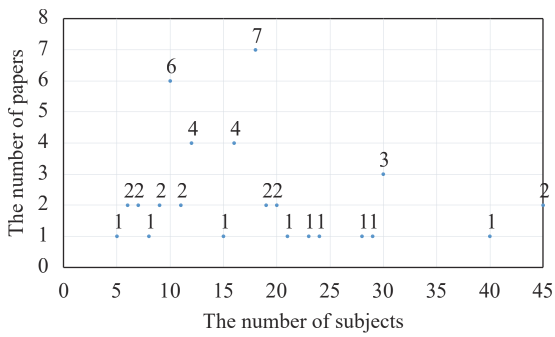


Fig. 4 The number of subjects used in different papers

The gender ratio is approximately balanced in most papers (with a difference of less than 1), but one study[42] chose to record only male data to avoid fluctuations within a monthly period. None of the research focuses exclusively on female participants, which may represent a gap to be filled.

3.3 EEG recordings

1) EEG devices and miniaturization trend

Fig. 5 lists the EEG devices used in all the research. Of all the devices, the most popular device is the BioSemi ActiveTwo system (including 64-channel and 128-channel versions), which accounts for 21.9% of all devices. In addition to the BioSemi system, the German product BrainAmp series was chosen by 12.5% of the researchers. The Neuroscan system and MINDWAVE Mobile are also popular, with 9.4% of researchers using their devices to conduct surveys on attention. In summary, the leading companies producing devices are from the United States (40.6%), the Netherlands (28.1%), and Germany (15.6%). There are also products from Italy, China, Spain, and other countries. More details about the EEG equipment can be found in Table 2.

The development trend of EEG wet electrode devices is toward more electrodes, with the number increasing from 32 to 128 in updated iterations. More electrodes provide better precision in EEG recordings, allowing researchers to obtain more detailed information about brain states using non-invasive devices[43].

There is also a growing trend toward miniaturization and portability of EEG devices[44], particularly in dry EEG devices. Wet electrode caps typically have 32, 64, or even 128 channels, making them inconvenient for real-world applications. To address this issue, several studies have focused on miniaturizing EEG equipment. Many portable EEG devices are mentioned in the literature collected for this review, including the Biosemi headset, the Nautilus mobile 16-channel EEG system, and the MindWave from NeuroSky. In-ear EEG has also become popular in recent years.

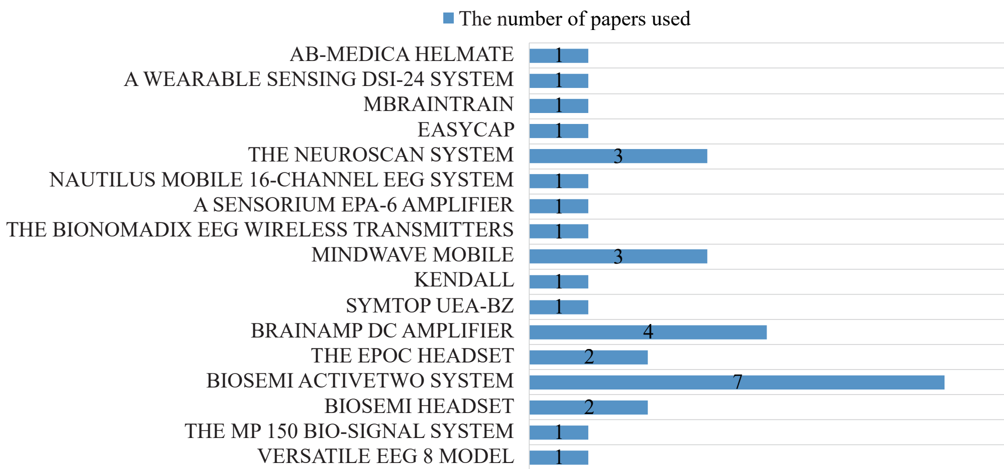


Fig. 5 EEG devices and the number of papers used

Table 2 The EEG device and their companies

Equipment model	Manufacturer
Versatile EEG 8 model	Produced by Bitbrain; Spain
The MP150 bio-signal instrumentation system	Produced by Biopac System Inc.; USA
Biosemi headset	Produced by BioSemi; Netherlands
Biosemi ActiveTwo system	Produced by BioSemi; Netherlands
The Epoc headset	Produced by Emotiv; USA
BrainAmp DC Amplifier	Produced by Brain Products GmbH; Germany
SYMTOP UEA-BZ	Produced by SYMTOP; China
Kendall	Produced by Covidien; USA
MindWave Mobile	Produced by NeuroSky; USA
The BioNomadix EEG wireless transmitters	Produced by Biopac Systems, Inc.; USA
A Sensorium EPA-6 amplifier	Produced by Sensorium Inc.; USA
Nautilus mobile 16-channel EEG system	Produced by Medical Engineering GmbH; Austria
The NeuroScan system	Produced by Compumedics Neuroscan; USA
Easycap	Produced by Easycap GmbH; Germany
mBrainTrain	Produced by mBrainTrain; Serbia
A Wearable Sensing DSI-24 system	Produced by Wearable Sensing; USA
AB-Medica Helmate	Produced by AB-Medica; Italy

The sampling rates reported are 128 Hz, 250 Hz, 256 Hz, 300 Hz, 500 Hz, 512 Hz, 1 000 Hz, 2 000 Hz, 2 048 Hz, and 8 192 Hz. The most frequently occurring frequency is 512 Hz, accounting for 24.4%, followed by 1 000 Hz at approximately 22.0%. The frequency of 8 192 Hz also has a high proportion of 9.8%, while the remaining frequencies have similar proportions. 58.5% of sampling rates are exact integer powers of 2, which is convenient for calculation. Moreover, 34.1% of the sampling rates are integer multiples of 100. More details are provided in Fig. 6.

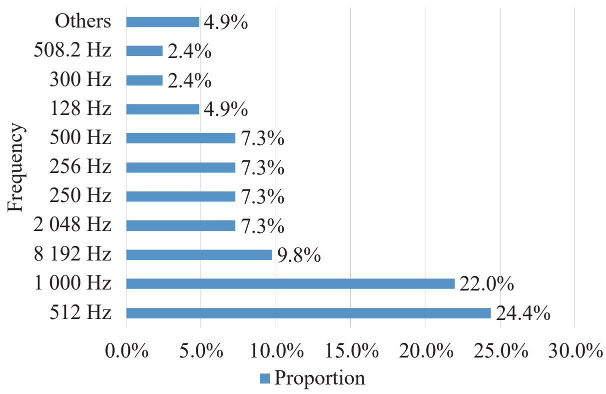


Fig. 6 The usage ratio of different sampling rates

2) Electrodes

Nearly all the studies reported the number of electrodes used. After analysis, it was found that the number of electrodes ranged from 1 to 128, with 64 being the most common. This is likely due to the widespread use of 64-channel electrode caps for data acquisition. It is the same case with the use of 16, 32, and 128. Recording data from all electrodes provides comprehensive information about brain activity and allows for flexible selection of channels during subsequent processing.

In addition to studies that record all electrodes, some researchers selectively record data from specific electrodes. As shown in Fig. 7, electrodes FP1 and FP2 have the highest frequency of use in attention-related studies. These electrodes are located in brain regions associated with attention, suggesting a close relationship between their activity and attentional processes. By selectively recording data from attention-relevant electrodes, researchers can avoid interference from irrelevant electrodes, reduce system complexity, and improve overall performance.

Some studies focus on miniaturizing and increasing the portability of EEG systems by using only one electrode[45]. Since the single electrode used must be strongly related to attention analysis, we mark the electrodes used in single-electrode studies with a star due to their importance.

3.4 Preprocessing

Preprocessing is essential for EEG data due to noise and artifacts. As shown in Fig. 8, of the 42 articles reviewed, only two used raw EEG data, both related to deep neural network (DNN). Band-pass filter is the most frequently used method, it removes irrelevant noise by retaining signals within a specific frequency range. The band-pass filter is commonly implemented by combining a high-pass filter with a low-pass filter. In this combina-

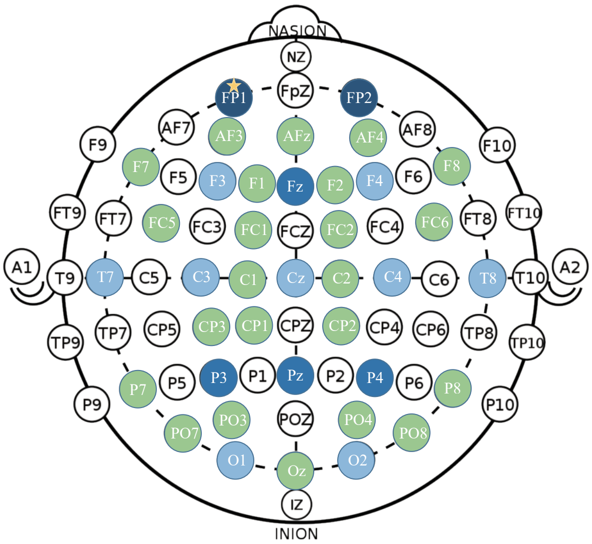


Fig. 7 Electrode usage frequency map, in which the marked electrodes appear in the following colors from high to low: dark blue, blue, light blue, and light green. The electrode marked with a star is the electrode used in the study of single-electrode classification problems. Electrodes that are not marked in color are not mentioned in the data collection process. (Colored figures are available in the online version at https://link.springer.com/journal/11633)

tion, the low-pass filter threshold typically ranges from 10 Hz to 70 Hz, with 50 Hz being the most prevalent value. High-pass filters are often set at 1 Hz, 0.5 Hz, and 2 Hz, while some studies opt not to apply a high-pass filter.

In addition to filtering, independent component analysis (ICA)[67] was frequently used to remove artifacts. ICA decomposes the EEG signal into sub signals using its inherent source separation principles, selectively eliminating components linked to eye movements, muscle activity, or environmental noise based on experiential discernment[68]. Other methods for removing noise and artifacts include artifact subspace reconstruction (ASR)[69], interpolation of bad channels, notch filtering, and linear methods.

MATLAB was the most popular tool for preprocessing, with 24 papers mentioning its use. The EEGLAB toolbox within MATLAB contains many algorithms for EEG preprocessing and visualization. Python is also a powerful tool for processing EEG data, with the MNE-Python toolkit providing capabilities for filtering and ICA.

3.5 Feature extraction

Feature extraction is an important step in EEG processing, as it can extract useful information from the time domain and frequency domain of EEG signals. Accurate analysis and interpretation of EEG data are crucial[70]. In this section, we introduce several common feature extraction methods and features that are used in attention analysis. The detailed results are listed in Table 3.

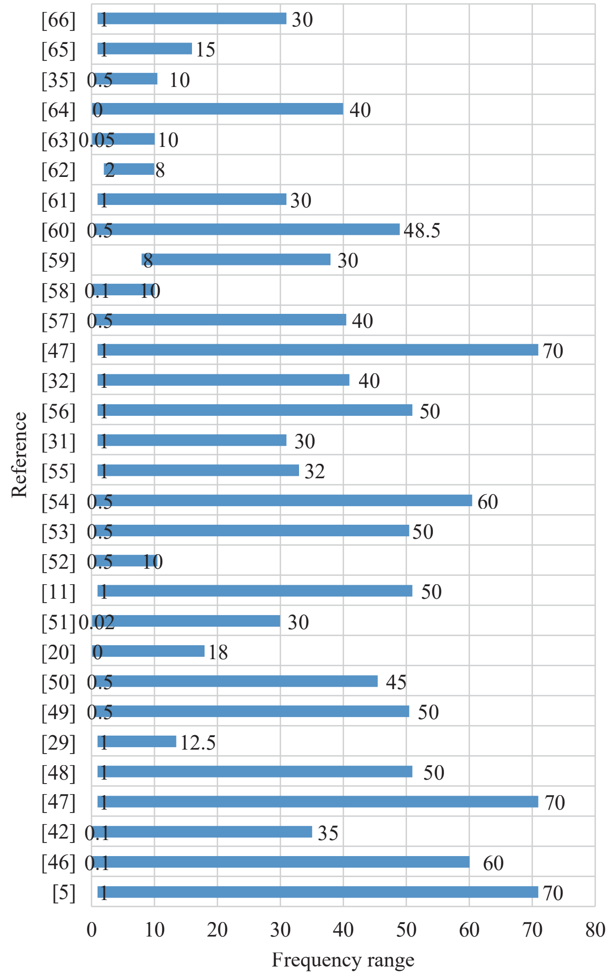


Fig. 8 Band-pass filter frequency range used in the listed articles

1) Feature extraction methods

According to [50, 84], feature extraction methods can be divided into several categories, including time domain analysis, frequency domain analysis, time-frequency domain analysis, and other methods such as spatial features, connectivity-related features and entropies. The number of articles that used methods from different categories is shown in Fig. 9.

Time domain analysis is based on the raw EEG signals or their statistical properties. Some common features that extract time domain information are event-related potentials (ERPs), Hjorth parameters, mean, standard deviation, kurtosis, skewness, variance, energy, and zero crossings. Frequency domain analysis reflects the spectral characteristics of EEG, such as the power and distribution of different frequency bands. The usual features are power spectral density (PSD), five frequency bands, fast Fourier transform (FFT), FFT coefficient, standard deviation, and differential entropy.

Time-frequency analysis recognizes characteristics from both the time domain and the frequency domain, it can capture the dynamic changes of EEG signals in both

Table 3 Analysis of preprocessing and feature extraction

Reference	Preprocessing	Feature extraction
[5]	Band-pass filter	–
[71]	Notch filter using FIR, band-pass filter, ICA	PSD, Hjorth parameters, entropy, kurtosis, PTP, zero crossings, skewness, Hurst exponent, mean, RMS, variance, SD, line length, quantile, decorrelation time
[42]	Band-pass filter, a disposable surface electrode	PSD, theta (4–8 Hz), SMR (12–15 Hz), middle beta (16–20 Hz)
[54]	Band-pass filter, linear method	FFT
[47]	Band-pass filter, reference (average), down-sampling (128 Hz)	3-D EEG features
[7]	Raw EEG	Spatio-temporal attention network
[29]	Band-pass filter	CCA
[46]	–	Theta (3–8 Hz), beta (12–30 Hz)
[49]	Band-pass filter	–
[30]	Raw EEG	3-D CNN
[20]	Band-pass filter	STFT (using Blackman window)
[51]	Reject bad channels, band-pass filter, ICA, down-sampling (64 Hz)	A sliding band-pass filter (2 Hz-window)
[53]	–	Algorithm optimization on the TGAM chip
[72]	ICA	–
[11]	Band-pass filter	STFT, PSD, delta, theta, alpha, beta, gamma, calculating the mean amplitude, standard deviation, peak-to-peak amplitude, skewness, and kurtosis for 0.5 s windows for each frequency band
[73]	Raw EEG	–
[56]	Band-pass filter, ICA	STFT, the average energy and power spectrum of rhythm waves
[74]	Band-pass filter, down-sampling (128 Hz)	Energy, entropy (including Shannon entropy, approximate entropy, sample entropy, and combined entropy), rhythmic energy ratio, frontal asymmetry ratio
[75]	A generic MWF-based removal algorithm	CNN
[57]	Band-pass filter	–
[58]	–	Filterbank common spatial pattern (FB-CSP) filtering
[31]	Reject bad channels, reference (average), band-pass filter, EEGLAB auto-reject, ICA	CSP
[59]	–	PSD (for alpha, beta, theta, gamma)
[76]	Band-pass filter, reference (average)	–
[61]	Multi-channel Wiener filter	Gammatone filterbank
[32]	Artifact subspace reconstruction (ASR)	Cross-correlation analysis
[33]	Band-pass filter	Different measures of brain connectivity (segregation, integration, and centrality)
[62]	Reject bad channels, ICA	ERP, time-frequency analysis for alpha (8.5–12 Hz) and theta (4–8 Hz) bands
[63]	Band-pass filter, ASR, reject bad channels and spherical interpolation of rejected channels	Mean P3 amplitude
[65]	–	Hilbert-Huang Transform (HHT), alpha, beta
[77]	Band-pass filter, ICA	CSP
[78]	Reject bad channels, band-pass filter, down-sampling (128 Hz)	Alpha band power, amplitude envelopes
[79]	Band-pass filter, ICA	Linear reconstruction model
[48]	Band-pass filter	Amplitude, time of peak negativity, slopes, variability, post-movement slope, Gaussian-windowed sinusoidal moving Morlet wavelet
[80]	ICA, high-pass filter	PSD, ERP
[35]	Band-pass filter, down-sampling (40 Hz)	Gammatone filterbank
[52]	Band-pass filter, down-sampling (125 Hz)	–

Table 3 (continued) Analysis of preprocessing and feature extraction

Reference	Preprocessing	Feature extraction
[55]	Band-pass filter, manual inspection	FFT
[81]	–	A linear encoding and decoding model
[82]	Band-pass filter, the joint decorrelation framework	Temporal envelope representations
[83]	Singular value decomposition (SVD)	FFT

Note: ICA, independent component analysis; PSD, power spectral density; PTP, peak-to-peak values; RMS, root mean square; SD, standard deviation; FFT, fast Fourier transform; CCA, canonical correlation analysis; CNN, convolutional neural network; STFT, short-time Fourier transform; "–" signifies that the corresponding aspect was not mentioned in the respective paper.

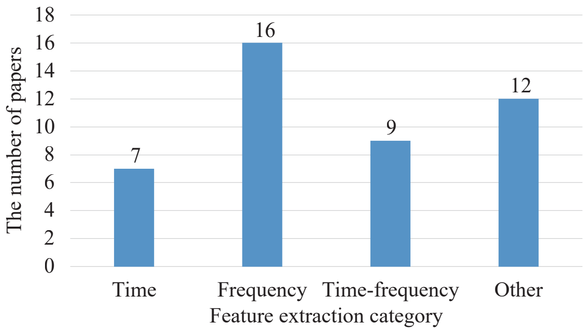


Fig. 9 The number of articles using the following category feature extraction methods: time domain, frequency domain, time-frequency domain, and others.

domains. Short-time Fourier transform (STFT), spectral entropies (SEs), discrete Fourier transform (DFT), and Hilbert transformation are all time-frequency features. Theta/Beta ratio (TBR) is a common hypothesis-driven feature in attention research. It has served as a valuable source for obtaining brain activity that may be associated with increased cognitive workload[64].

In addition to common time or frequency domain features, other methods for feature extraction include spatial features, connectivity-related features and entropies[50]. Spatial feature extraction involves techniques such as common spatial pattern (CSP) and filter-bank common spatial pattern (FB-CSP) filtering[66]. Connectivity-related features in EEG, derived from pairwise interactions and complex networks, capture the broader network dynamics of interactions among multiple channels. These features, such as functional connectivity, offer insights into the collective patterns of EEG signal connectivity. Entropies, such as Shannon entropy and approximate entropy, quantify uncertainty and randomness in observed EEG data.

However, there are no fixed rules for choosing the best EEG features, as most feature extraction methods are domain knowledge-based and are hypothesis-driven data modelling outcomes. These data serve as input references for subsequent analysis, mainly traditional machine learning methods[85]. For example, Fan et al.[74] used 97 features to achieve a comprehensive and thorough analysis of EEG signals.

2) Feature selection methods

The purpose of feature selection is to choose a subset of the available features, to improve computational efficiency and avoid the curse of dimensionality[84]. For example, Fan et al.[74] employed CatBoost feature selection (CatB-FS) to reduce 97 features to the most effective combination, while Geravanchizadeh and Gavgani[33] used a k-nearest neighbors (KNN) classifier for feature selection. In another study, neural networks (NNs) were utilized for feature selection[54]. Additionally, feature averaging is a common data reduction method, which involves calculating the mean value of features across trials or subjects to mitigate variability.

3.6 Classification

This section provides an overview of the most widely applied analysis algorithms for EEG signals and evaluates their strengths and weaknesses. We categorize these methods as traditional machine learning methods and deep learning methods. Table 4 lists the traditional classification methods we surveyed, and Table 5 summarizes the deep learning methods used in the literature we surveyed. Traditional machine learning methods rely on manually designed features and rules, and learn relationships between data through models. However, their effectiveness is limited when dealing with complex data and abstract concepts. In contrast, deep learning methods automatically learn hierarchical representations of data through multi-layered neural networks. These methods excel at extracting abstract features from large-scale datasets, making them suitable for handling complex and high-dimensional information.

1) Traditional machine learning methods

Traditional machine learning methods are advantageous for high-dimensional, nonlinear signal classification capabilities, which are often required in EEG analysis. About 48.8% of the articles used traditional machine learning methods for classification, with the most popular being support vector machine (SVM) (13.6%), linear discriminant analysis (LDA) (11.36%), KNN (9.09%), random forest (RF) classifier (6.82%) and convolutional neural network (CNN) (6.82%). Other methods include canonical correlation analysis (CCA), light gradient boosting machine (LightGBM), extreme gradient boosting (XGBoost), decision tree (DT), stacking cross validation (StackingCV), and DNN.

Statistical tests are also widely used in EEG signal

Table 4 Analysis of the works using traditional machine learning methods

Reference	Online VS. Offline	Is cross-subject	Classifier	Evaluation metric
[71]	–	Y	SVM, k-NN, RF	Acc: 88.9% (binary model), 81.9% (multiclass model)
[42]	–	Y	The ratio of different waves ([SMR + middle beta]/theta)	–
[54]	–	Y	NNs, multivariate linear regression	R^2, adjusted R^2, RMSE, Pearson's linear correlation coefficient
[29]	Online	Y	CCA	Acc: 88.5% (SD: 7.8%), the average information transfer rate: 3.02 bit/min
[46]	Online	N	The TBR method	–
[20]	Offline	Both	SVM(5-fold), k-NN, ANFIS	Acc: 96.7% (subject-specific), 78.8% (common subject)
[51]	Offline	Y	Multivariate linear reconstruction model + LDA	Acc: 93%
[56]	–	–	LSTM, RF, CNN	–
[74]	Offline	N	SVM, k-NN, DT, RF, LightGBM, XGBoost, StackingCV, a time-series voting algorithm	Recall: 89.44%, precision: 93.48%
[57]	Offline	Y	LDA, CNN	Acc: 90.6% (SD: 10% lower-dimensional datasets), 90.7% (SD: 8.6% higher-dimensional datasets)
[58]	Offline	Both	LDA	Acc: 80% (1s-window)
[31]	Offline	Both	SVM	Acc: 80%–83%
[59]	Offline	–	LDA	Acc: 85.37% ± 11.27%
[76]	Offline	Y	A linear model based on least squares error minimization	Acc: 82.7% (NH listeners), 59.79% (CI users)
[61]	Offline	Y	A linear decoder	Spearman correlation between the real and reconstructed speech envelope
[32]	Online	Y	An Online AAD processing pipeline	–
[33]	–	Y	k-NN, SVM, DNN, CNN	Acc: 89.5%
[54]	Offline	Y	MLR, MVPA	R^2, adjusted R^2, RMSE, Pearson's linear correlation coefficient
[62]	Offline	Y	SVM	Acc: 50%–85%
[63]	Online	Y	LMM	–
[65]	Offline	Y	SVM	Acc: 84.80%
[34]	–	–	A linear stimulus-reconstruction decoder, a neural-network stimulus-reconstruction decoder, CNN, DNN	Acc: 81% (wet EEG), 87% (dry EEG)
[77]	Offline	N	k-NN, SVM, ANN, LDA, NB	Acc: 92.8% ± 1.6%
[78]	Offline	Y	Regularized linear regression, multivariate backward modelling, SVM	Acc: 85.7% (moving talker), 77.8% (stationary talker)
[79]	Offline	Y	Decoding selective attention with dichotic stimulation	Acc: 80.9% (NH listeners), 71.5% (CI users)
[48]	Offline	Y	LDA, PCA	Acc: 71% ± 10.9%
[52]	Offline	Y	Ridge regression to train forward encoding models	Acc: 77%
[86]	Online	–	The state-space model based on Bayesian fixed-lag smoothing	–
[83]	Offline	Y	APN model	Acc: 90.4%, precision rate: 92.12%, recall rate: 84.68%

Note: SVM, support vector machine; k-NN, k-nearest neighbors; RF, random forest classifier; acc, accuracy; SMR, sensorimotor rhythm; NNs, neural networks; RMSE, root-mean-squared error; CCA, canonical correlation analysis; TBR, theta-to-beta ratio; ANFIS, adaptive neuro-fuzzy inference system; LDA, linear discriminant analysis; CNN, convolutional neural network; SD, standard deviation; LSTM, long short-term memory; DT, decision tree; NH listeners, normal hearing listeners; CI users, cochlear implant users; AAD, auditory attention detection; DNN, deep neural network; MLR, multiple linear regression; MVPA, multivariate pattern analysis; LMM, linear mixed model; ANN, artificial neural network; NB, naive Bayes; PCA, principal component analysis; APN, associative petri net. "Online VS. Offline" refers to whether the study involves real-time Online processing or Offline analysis. "is cross-subject" indicates whether the study analyzes data from the same individuals or different individuals, respectively. "–" signifies that the corresponding aspect was not mentioned in the respective paper.

analysis, as they are effective and easy to interpret. Statistical tests are the process of applying mathematical models to test hypotheses or compare the differences between groups or conditions. Common statistical tests include analysis of variance (ANOVA), Pearson's correlation, and linear mixed models (LMM).

Table 5 Analysis of the works using deep learning methods

Reference	Layers	Optimization	Loss function + training strategy	Hyperparameter tuning	EEG related design	Evaluation metric
[5]	3 Stacked LSTM layers	Adam	Weighted cross entropy	–	Capsule att	RMSE, PCC
[47]	1 conv + 1 max-pooling + 2 FC	Random gradient descent	Cross_entropy + learning rate decay	–	Frequency att + channel att	Acc: 83.6% (1s-window), 86.9% (2s-window)
[7]	1 conv + 2 FC + 1 self-att + 1 FC	Adam	–	Grid-search	Spatial att + temporal att	Acc: 90.1% (1s-window, KUL), 71.9% (1s-window, DTU)
[30]	1 Input + 3 Block + 1 flatten + 3 FC	Adam	Cross_entropy + learning rate decay	–	3D conv + 3D presentation of EEG	F1: 0.80
[72]	3 conv + 3 ResNet + 5 conv + 1 GRU	–	Dropout = 0.3, batch normalization	–	STIN	F1: 0.92 (2 class), 0.88 (task-specific)
[11]	Input layer + reservoir layer + output layer	–	–	Grid-search	ESN	Acc: 81.16%
[73]	2 LSTM layers	Adam	–	Early stopping	BLSTM-RNN	SINR, PESQ
[75]	1 conv + 1 average-pooling + 2 FC	–	–	Grid-search	–	Acc: 80.8% (1s-window), MESD: 0.819s

Note: LSTM, long short-term memory; att, attention; RMSE, root mean square error; PCC, pearson correlation coefficient; conv, convolution layer; FC, fully-connected; acc, accuracy; self-att, self-attention; KUL, KUL dataset; DTU, DTU dataset; F1, F1 score; GRU, gated recurrent unit; STIN, spatial temporal information network; ESN, echo state network; BLSTM-RNN, bidirectional long short-term memory recurrent neural network; SINR, signal-to-interference-plus-noise ratio; PESQ, perceptual evaluation of speech quality; MESD, minimal expected switch duration; "–" signifies that the corresponding aspect was not mentioned in the respective paper.

However, these methods may not be able to simultaneously capture the time-domain, frequency-domain, or spatial features of EEG signals. Some methods based on linear assumptions also struggle to handle the non-linear characteristics of EEG signals. In subsequent experiments, we will observe that methods such as LDA perform less effectively across all datasets than deep learning approaches.

2) Deep learning methods

With the development of deep learning methods, many articles in the domain of EEG research have applied deep learning models to analyse EEG signals. In this section, we summarize in detail the common deep learning model structure, training strategy, and improvements made for attention analysis in EEG.

The deep learning model designed for attention detection is based on several basic models, with certain adjustments to the structure or input. The basic structures of these models are CNN, LSTM, RNN, and DNN. Table 5 lists the specific layer structures of the different models in the literature of the past five years. As shown, convolutional layers are the most common among all the structures condsidered, indicating that the feature extraction principle of convolutional layers is well suited for extracting attention information from EEG signals. Other model layers that appear include LSTM layers, capsule attention layers, pooling layers, self-attention mechanisms, cascading blocks, residual blocks, gated recurrent unit (GRU) layers, and flattening layers.

The training strategy is also an important aspect of deep learning models. Among all the literatures that use deep learning models for classification, none of them use pre-trained models. We will discuss this topic from loss function, optimization, regularization, and parameter search.

The loss function can quantify the error between the model prediction and the ground truth. The most commonly employed loss functions include the mean squared error loss function, weighted cross-entropy loss function, and binary cross-entropy loss function. To achieve better training results, some studies have used gradient descent and adaptive learning rate reduction for training. For example, in [30], the initial training learning rate is set to 0.001 and multiplied by 0.4 after every 10 rounds.

The optimizer is an algorithm that updates the model parameters based on the loss function and the gradient. The most frequently used optimizer is adaptive moment estimation (ADAM)[60]. ADAM can achieve fast convergence and robustness to noise and sparse gradients. No other optimizer was mentioned in these articles.

The regularization methods used include drop-out and batch normalization, with [72] mentioning a drop-out setting of 0.3, and the mentioned batch size is 32 and 1 024. It can help prevent overfitting and improve the generalization performance of deep learning models. The rest literatures do not mention regularization methods.

Parameter search is a process of finding the optimal values for the hyperparameters of a deep learning model, such as the learning rate or the number of layers. It can improve the performance and efficiency of deep learning models. Jeong et al.[11] use a grid search to perform parameter search, which is a method that exhaustively tries all possible combinations of values within a predefined range or set.

All the literatures using deep learning methods have made targeted improvements to the model based on the characteristics of EEG. Zhang and Etemad[5] used a capsule attention mechanism to learn the hierarchical relationship in the input data to better handle the noise and artifacts in EEG data. Cai et al.[47] added frequency attention and channel attention to optimize model performance. STAnet[7] utilized spatial and temporal attention mechanisms to extract information from EEG signals. The spatial attention mechanism assigns varying weights to EEG channels dynamically, while the temporal attention mechanism captures temporal patterns in the signals. Zhang et al.[30] converted 28 EEG electrodes into a 9×9 spatial matrix to preserve the spatial topological relationship between electrodes and adopted a 3D convolutional neural network model that can simultaneously explore spatial and temporal information. Li et al.[72] proposed a temporal-spatial information network to extract temporal and spatial features of EEG signals.

3) Evaluation

Baseline: To evaluate the performance of the proposed method, some paper compared their models to the baseline model. The usual baseline model includes CNN, long short-term memory (LSTM) network, CNN-LSTM, and DNN. There are also proposed methods used as a baseline, such as EEGNet, DeepNet, CapsNet, and CCA.

Metric: To evaluate the performance of different machine learning models for EEG classification, various metrics have been used in these articles. The most common metrics are accuracy (acc) and the F1 score. Other metrics that have been used to evaluate machine learning models for EEG classification include root-mean-squared error (RMSE), the information transfer rate (ITR), R2, adjusted R2, Pearson correlation coefficient (PCC), recall rate, computation speed, etc. The number of classes, the machine learning model used, the subject-specificity, and the length of the time window, will all influence the model′s performance. For example, Fan et al.[74] reported an accuracy of 81.9% for multiclass models, while it increased to 88.9% for binary models. These metrics can provide additional insights into the performance and reliability of machine learning models for EEG classification.

Online VS. Offline: In the literatures we surveyed, the majority (68.9%) of the researchers studied only the offline problem, 16.0% of the papers focused on the performance of the classifier in online systems, and 16.0% of the papers did not mention whether the classifier was online or offline. According to the classification performance, offline-trained classifiers applied to online systems have lower accuracy in online systems than in offline states. First, the experimental paradigm and the real-world task have great differences, which makes the offline classifier perform worse in online systems. Second, many online system classifiers are offline-trained and do not adapt to the use of online systems[87]. Because feedback is also an important part of attention-related human-computer interaction systems, such as attention enhancement problems, it is necessary to study the performance of classifiers in online systems.

Within-subject VS. Cross-subject: Due to the influence of various factors such as environmental noise or the subject′s psychological state, the EEG signals collected from different subjects vary greatly[88]. Therefore, cross-subject is an important issue in the EEG field. Studying cross-subject classification methods can improve the robustness of EEG recognition by maximizing the correlation between and within subjects[89]. We collected and analysed the information on whether the literature involved cross-subject issues, and found that 65.5% of the articles claimed that their methods were applicable to cross-subject situations, 10.3% of the researchers reported both cross-subject and within-subject classifier performance, 10.3% of the researchers only studied within-subject situations, and 13.7% of the articles did not specify whether their methods were cross-subject or within-subject. We found that most articles tested classifier performance across subjects, and that the same classifier usually performed significantly better in within-subject situations than in cross-subject situations. To address this issue, there are several possible strategies: 1) Train subject-specific models by individually training models for different subjects. 2) Implement subject-specific feature alignment to mitigate issues related to misalignment in subject data domains. 3) Introduce data augmentation techniques to enhance the model′s generalization capabilities. 4) The choice of model architectures, such as network depth and neuron count, may impact the model′s generalizability capabilities[90].

4) Experimental results

Datasets: To evaluate and compare different model performances from the common dataset, we conducted experiments using three diverse datasets to cover different aspects of attention analysis: one[30] for attention state classification, another[31] for mind wandering detection, and the KUL dataset for AAD issues[27] (Table 1 provides detailed descriptions). We employed the same preprocessing methods as those outlined in the original papers.

Classification models: As the deep learning codes from the reviewed papers were not open-sourced, we reproduced two representative models based on the following architectures: STAnet[7] for AAD and 3D-CNN[30] for attention state classification. We also conducted experiments on the three datasets. The reproduced STAnet achieved 89.45% accuracy on the KUL dataset, slightly lower than the reported 90.1% accuracy reported in the original paper. Conversely, the reproduced 3D-CNN obtained 81.69% accuracy on the Zhang et al. dataset[30], surpassing the 70.15% accuracy reported in the original paper. The lack of specific information on the dropout layers in the original paper may have contributed to the difference in performance of the 3D-CNN model in the reproduction. The source code for the deep learning models reproduced in this paper is available on GitHub at https://github.com/christine-daae/DLModels.

For traditional machine learning methods, we employed the widely-used PSD, FFT, and SEs methods for feature extraction, and in the classification step, we experimented with three commonly-used traditional machine learning methods (SVM, LDA, KNN) across all the datasets. The primary metric used for model performance evaluation was accuracy.

Experiment setup: The experiments were performed using a Nvidia GeForce 2080Ti. The 3D-CNN model was implemented in TensorFlow 2.0, while STAnet was implemented in PyTorch. Parameters were set consistent with the original papers, and a dropout rate of 0.5 was uniformly applied to prevent overfitting. The training batch size was set to 16, and the models were trained for 30 epochs.

Results: All the experimental results are shown in Table 6. The KNN classifier with FFT outperformed, achieving a notable accuracy of 86.23% on the Zhang et al. dataset[30]. Meanwhile, STAnet demonstrated superior performance with accuracies of 67.38% on the Dhindsa et al. dataset[31] and 89.45% on the KUL dataset[27]. Overall, STAnet performs the best on average among all classifiers.

Table 6 Attention detection accuracy (%) of different classification methods on three datasets

Model	Zhang et al.[30]	Dhindsa et al.[31]	KUL[27]
STAnet	84.40	**67.38**	**89.45**
3D-CNN	81.69	65.70	72.70
SVM + PSD	69.53	65.76	50.59
SVM + SEs	74.50	65.57	60.04
SVM + FFT	71.00	66.28	77.87
LDA + PSD	76.50	61.50	74.62
LDA + SEs	74.84	65.28	71.87
LDA + FFT	73.15	58.21	65.56
KNN + PSD	70.69	53.28	59.62
KNN + SEs	73.65	51.50	65.06
KNN + FFT	**86.23**	53.78	65.12

Note: PSD, power spectral density; SEs, spectral entropies; FFT, fast Fourier transform; SVM, support vector machine; LDA, linear discriminant analysis; k-NN, k-nearest neighbors.

Deep learning models generally demonstrated greater stability and superior performance. Among them, STAnet consistently outperforms 3D-CNN across all three datasets, which may be attributed to its distinctive network structures. The complexity of STAnet, which involves feature extraction in both temporal and spatial dimensions along with the use of attention mechanisms for global feature extraction, contributes to its superior expressive power and overall better performance. In contrast, the relatively simplistic 3D-CNN network structure relies solely on a series of 3D convolutions for feature extraction.

Traditional machine learning methods are more dependent on feature extraction and selection. When selecting suitable feature combinations, the performance of traditional machine learning methods may match or even surpass that of deep learning models. However, achieving this requires deliberate manual selection and experimentation with various feature combinations. Furthermore, while identical feature combinations may perform well on one dataset, their performance may be more modest on other datasets.

In summary, deep learning methods demonstrate notable accuracy and stability without the need for explicit feature selection. On the other hand, traditional machine learning classification methods can achieve high accuracy through careful feature selection, while maintaining faster computation and training speeds than deep learning models. On the Zhang et al. dataset[30], traditional machine learning and deep learning models exhibit varied performances. Deep learning significantly outperforms other methods on the KUL dataset[27]. For the Dhindsa et al. dataset[31], 3D-CNN and SVM showed minimal performance differences. Concerning the poor performance of all classifiers on the Dhindsa et al. dataset[31], we suspect that the subjective judgments of participants regarding mind-wandering, used as labels in this dataset, might introduce errors and inconsistencies in the labeling process, thereby explaining the subpar classification performance.

4 Discussions

4.1 Best practice recommendations

In this section, we present some practical recommendations based on previous research results. Here are the detailed suggestions.

1) Data acquisition

Specify the EEG equipment and the sampling rate used in the experiment.

• Use as many subjects as possible. The number of subjects should not be less than 20 preferably. The gender ratio of subjects should also be balanced.

• Use a well-established experimental paradigm to ensure the reliability of the collected data.

• Minimize environmental interference as much as possible. A comfortable environment can reduce eye movement and body movemen, and help individuals concentrate at the same time.

• Specify the electrode distribution standard of the recording equipment (such as the 10–20 system).

• List the electrodes used in the article.

2) Artifacts handling

• Indicate the frequency range of the band-pass filter.

- List the artifact handling methods used.
- Give the specific version of the toolkit used.
- Explain the reason for keeping specific artifacts. For example, document A retains eye movement signals because they believe that they can be used as a reference for attention status.

3) Feature extraction

- Present all the feature extraction methods and provide detailed parameter settings.
- Summarize each method's principle and formula, and compare their performance using tables or figures.
- Elucidate the correlation mechanism between these features and the attentive state. If there is a feature selection method, describe its principle and effect as well. If there are relevant theories or experiments to support them, cite the literature.
- If the article did not use feature extraction methods, briefly explain the reason (such as features are directly extracted by a 3-D convolutional network).

4) Classification

- Give the implementation details of the classifier, such as SVM kernel function selection.
- If the article does something special with categorical data (e.g., split into 10 folds), list them explicitly.
- When using deep learning methods, explain the specific model structure (it will be more intuitive to have a model structure diagram).
- Present training strategy, optimization function, and other settings of the model in a table or a paragraph.
- Use various evaluation metrics to omnidirectionally show the performance of the model.
- List the metrics used and their values.
- Give a clear explanation of the experimental results and, if there is unusual data, give possible reasons.

4.2 Open problems

Despite the progress made in the field of attention detection using EEG signals, there are still many open problems that need to be solved. We summarize some of them as follows:

1) Need for high-quality data

Obtaining high-quality EEG data for attention detection is challenging due to various factors, such as privacy and ethical issues, diverse experimental paradigms, vague definitions of attention and its subtypes, and a lack of public datasets. In addition, many studies use their own collected data with a small number of subjects, which limits the reproducibility and generalizability of their results. Moreover, it is difficult to control all the variables during the data collection process. The state of the subjects such as fatigue or boredom, and environmental changes such as temperature or background noise can impact data quality.

2) Lack of standardization and generalization

There is no unified and convincing standard for measuring the degree of attention or classifying different attention states. Besides, most of the studies on EEG-based attention detection use different tasks, stimuli, data processing, and evaluation methods, which make it difficult to compare and evaluate different methods and results across studies. Moreover, there is also a lack of standardized evaluation metrics available for achieving fair comparisons between various studies.

3) Low spatial resolution and limited electrode locations in portable devices

EEG has a low spatial resolution compared to other neuroimaging techniques. This limitation is partly influenced by hardware constraints, and electrode selection is partly guided by the known attention processing pathways. Moreover, the small and wearable devices restrict the number and position of electrodes that can be used to measure EEG signals. Therefore, finding the optimal electrode configuration and trade-off between spatial sampling and temporal resolution is a challenge.

4) Need for advanced analysis methods

Current classification methods have several limitations: 1) There is a trade-off between accuracy and efficiency, as longer decision windows may improve accuracy but increase delay time[7]. 2) The performance drops in cross-subject tasks, as individual differences may affect the generalization ability of the models. 3) The performance decreases in online tasks, when the offline trained model is applied to the online system, the performance of the model will also decrease to varying degrees. 4) Only a few studies have used deep learning methods and improved the model structure for the attention problem. 5) Many papers did not make their datasets and code public, or provide details about model parameters, which limits the reproducibility of their results. Therefore, developing more accurate, robust, interpretable, and efficient machine learning algorithms for EEG-based attention detection is necessary.

5) Need for real-world scenarios

Most studies are based on highly controlled and artificial scenarios that do not reflect real-world situations where attention detection may be useful. In real-life scenarios, there may be more background noise, interference factors, or unexpected events that may affect the EEG signals and the attention state. Besides, nearly half of the researchers use more than 20 electrodes to record EEG signals, which is impractical and inconvenient for real-world usage. Therefore, we need to study how to use fewer electrodes or other devices to achieve better classification accuracy and improve the practical application value of the method.

4.3 Future trend

We summarize the views of these articles and propose the following trends which we believe are valuable.

1) More public dataset

We noticed that compared to other EEG study fields, the public dataset in the attention field is relatively small, and is a blank to be filled. Over half of the studies used self-collected data, making it hard to reproduce and validate their work. Besides, 25.7% of studies use fewer than 10 participants, the more high-quality public datasets may help them perform additional experiments to increase the reliability of their argument. Thus, more datasets and various subjects are needed.

2) Traditional machine learning methods

Traditional machine learning methods continue to be integral in advancing EEG-based attention detection. Here are several potential future trends in this domain: i) Explore ensemble learning techniques to combine strengths from different traditional machine learning models, such as random forests or boosting. ii) Progress in feature engineering techniques aims to improve the representation of EEG data. Researchers are exploring ways to extract relevant features from EEG signals, aiming to optimize information input into classical machine learning models. iii) Develop methods to explain decisions made by these models in the context of attention detection, promoting transparency and interpretability. iv) Adapt to real-world applications, where attention states may dynamically change over time.

3) Deep learning methods

There are several possible trends for deep learning methods in attention detection using EEG signals:

i) Apply novel deep learning architectures, such as CNN or attention mechanisms, to model the spatial and temporal dynamics of EEG signals and attention processes.

ii) Explore the use of generative models, like generative adversarial networks, to synthesize realistic EEG signals or stimuli for attention detection or modulation.

iii) Investigate the interpretability of deep learning models for EEG-based attention detection to improve the trustworthiness and transparency of the systems.

iv) Integrate deep learning techniques with other machine learning or signal processing methods to enhance the performance and efficiency of EEG-based attention detection.

v) Optimize the trade-off between spatial sampling and temporal resolution in EEG-based attention detection to improve wearability and comfort.

4) Expanding sensor selection integrated with cognitive theories

In future research, we anticipate a trend toward deeper integration of sensor selection with cognitive theories. With continuous advancements in hardware technology, we can expect more flexible and diverse sensor layouts in EEG studies. This trend will contribute to a more comprehensive understanding of neural processes related to attention. Furthermore, we foresee a more in-depth exploration of high-dimensional data to drive data-driven exploration. As technology progresses, the ability to handle larger and more complex EEG datasets will deepen our understanding of attention research.

5) Real-world scenarios

The existing studies are usually based on simple manual scenarios and do not perform well in practical applications. In real-world environments, there might be more interference or accidents like damaged electrodes. In auditory attention detection, there are many complex scenes such as speech mixtures, overlapping sounds, moving speakers, or background noise. More research is needed to investigate the neural mechanisms and dynamics of human attention levels in complex and realistic scenarios.

6) Development of neuro-steered applications

EEG-based attention detection can be used for developing neuro-steered applications, such as hearing aids and auditory prostheses. These devices can adapt to the user's focus of attention and enhance the desired sound source. Moreover, EEG-based attention detection can be used for developing brain-computer interfaces that can monitor and modulate the user's attention level and provide real-time feedback or intervention. Therefore, an increasing number of researchers are studying the feasibility and effectiveness of neuro-steered applications using EEG-based attention detection.

7) Personalized and adaptive EEG-based attention detection

Attention patterns can vary widely across people and contexts, and they can also change over time due to factors such as mind wandering, distraction, and fatigue. These variations and changes can affect the performance and behavior of the user. Therefore, it is important to develop personalized and adaptive models for attention detection using EEG signals, these models can capture individual differences and fluctuations in attention states and provide appropriate feedback or intervention.

8) Multimodal data

In recent years, researchers have been incorporating multiple physiological signals to obtain more precise measurements of mental states. Eye movements, muscle activity, and facial expressions are all valuable references for evaluating attention. The multimodal data can provide complementary and redundant information that can improve the accuracy and robustness of EEG-based attention detection. Therefore, more research is needed to explore the use of multimodal data for EEG-based attention detection.

5 Conclusions

In this article, we systematically and comprehensively review the research on attention in the field of EEG over the past 5 years. We mainly focus on two issues: AAD and attention level classification. We summarize the mainstream EEG data preprocessing and feature extraction methods and provide a detailed analysis of machine learning methods found in the literature. We also consider the factors that influence the effectiveness and suitability of these models. Furthermore, we offer some suggestions for data acquisition, feature extraction, and classification methods. Our review also reveals some existing problems and challenges in this field, such as the lack of standardized datasets, the trade-off between accuracy and efficiency, the ethical and privacy issues of EEG data collection and analysis, etc. Therefore, we suggest some possible future research directions such as using multimodal data to improve the accuracy and robustness of attention detection. We hope that our research results can provide an objective and useful practical guide for future researchers.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (Nos. 62136004, 62276130 and 62406131), the National Key R&D Program of China (No. 2023YFF1204803), and Key Research and Development Plan of Jiangsu Province, China (No. BE2022842).

Declarations of Conflict of interest

Daoqiang Zhang is an Associate Editor for *Machine Intelligence Research* and was not involved in the editorial review, or the decision to publish this article. All authors declare that there are no other competing interests.

References

[1] M. M. Chun, J. D. Golomb, N. B. Turk-Browne. A taxonomy of external and internal attention. *Annual Review of Psychology*, vol. 62, pp. 73–101, 2011. DOI: 10.1146/annurev.psych.093008.100427.

[2] R. H. C. E. Souza, E. L. M. Naves. Attention detection in virtual environments using EEG signals: A scoping review. *Frontiers in Physiology*, vol. 12, Article number 727840, 2021. DOI: 10.3389/fphys.2021.727840.

[3] A. Gaume, G. Dreyfus, F. B. Vialatte. A cognitive brain-computer interface monitoring sustained attentional variations during a continuous task. *Cognitive Neurodynamics*, vol. 13, no. 3, pp. 257–269, 2019. DOI: 10.1007/s11571-019-09521-4.

[4] F. Aoki, E. E. Fetz, L. Shupe, E. Lettich, G. A. Ojemann. Increased gamma-range activity in human sensorimotor cortex during performance of visuomotor tasks. *Clinical Neurophysiology*, vol. 110, no. 3, pp. 524–537, 1999. DOI: 10.1016/S1388-2457(98)00064-9.

[5] G. Y. Zhang, A. Etemad. Capsule attention for multimodal EEG-EOG representation learning with application to driver vigilance estimation. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 29, pp. 1138–1149, 2021. DOI: 10.1109/TNSRE.2021.3089594.

[6] A. M. Narayanan, A. Bertrand. Analysis of miniaturization effects and channel selection strategies for EEG sensor networks with application to auditory attention detection. *IEEE Transactions on Biomedical Engineering*, vol. 67, no. 1, pp. 234–244, 2020. DOI: 10.1109/TBME.2019.2911728.

[7] E. Z. Su, S. Q. Cai, L. H. Xie, H. Z. Li, T. Schultz. STAnet: A spatiotemporal attention network for decoding auditory spatial attention from EEG. *IEEE Transactions on Biomedical Engineering*, vol. 69, no. 7, pp. 2233–2242, 2022. DOI: 10.1109/TBME.2022.3140246.

[8] C. Reichert, S. Dürschmid, M. V. Bartsch, J. M. Hopf, H. J. Heinze, H. Hinrichs. Decoding the covert shift of spatial attention from electroencephalographic signals permits reliable control of a brain-computer interface. *Journal of Neural Engineering*, vol. 17, no. 5, Article number 056012, 2020. DOI: 10.1088/1741-2552/abb692.

[9] P. Miranda, C. D. Cox, M. Alexander, S. Danev, J. R. Lakey. In quest of pathognomonic/endophenotypic markers of attention deficit hyperactivity disorder (ADHD): Potential of EEG-based frequency analysis and ERPs to better detect, prevent and manage ADHD. *Medical Devices: Evidence and Research*, vol. 13, pp. 115–137, 2020. DOI: 10.2147/MDER.S241205.

[10] J. A. O'Sullivan, A. J. Power, N. Mesgarani, S. Rajaram, J. J. Foxe, B. G. Shinn-Cunningham, M. Slaney, S. A. Shamma, E. C. Lalor. Attentional selection in a cocktail party environment can be decoded from single-trial EEG. *Cerebral Cortex*, vol. 25, no. 7, pp. 1697–1706, 2015. DOI: 10.1093/cercor/bht355.

[11] D. H. Jeong, J. Jeong. In-ear EEG based attention state classification using echo state network. *Brain Sciences*, vol. 10, no. 6, Article number 321, 2020. DOI: 10.3390/brainsci10060321.

[12] S. M. Alarcão, M. J. Fonseca. Emotions recognition using EEG signals: A survey. *IEEE Transactions on Affective Computing*, vol. 10, no. 3, pp. 374–393, 2019. DOI: 10.1109/TAFFC.2017.2714671.

[13] M. Adamou, T. Fullen, S. L. Jones. EEG for diagnosis of adult ADHD: A systematic review with narrative analysis. *Frontiers in Psychiatry*, vol. 11, Article number 871, 2020. DOI: 10.3389/fpsyt.2020.00871.

[14] H. Altaheri, G. Muhammad, M. Alsulaiman, S. U. Amin, G. A. Altuwaijri, W. Abdul, M. A. Bencherif, M. Faisal. Deep learning techniques for classification of electroencephalogram (EEG) motor imagery (MI) signals: A review. *Neural Computing and Applications*, vol. 35, no. 20, pp. 14681–14722, 2023. DOI: 10.1007/s00521-021-06352-5.

[15] J. Belo, M. Clerc, D. Schön. EEG-based auditory attention detection and its possible future applications for passive BCI. *Frontiers in Computer Science*, vol. 3, Article number 661178, 2021. DOI: 10.3389/fcomp.2021.661178.

[16] E. Alickovic, T. Lunner, F. Gustafsson, L. Ljung. A tutorial on auditory attention identification methods. *Frontiers in Neuroscience*, vol. 13, Article number 153, 2019. DOI: 10.3389/fnins.2019.00153.

[17] G. A. M. Vasiljevic, L. C. De Miranda. Brain–computer interface games based on consumer-grade EEG Devices: A systematic literature review. *International Journal of Human–Computer Interaction*, vol. 36, no. 2, pp. 105–142, 2020. DOI: 10.1080/10447318.2019.1612213.

[18] N. M. Yusoff, R. F. Ahmad, C. Guillet, A. S. Malik, N. M. Saad, F. Mérienne. Selection of measurement method for detection of driver visual cognitive distraction: A review. *IEEE Access*, vol. 5, pp. 22844–22854, 2017. DOI: 10.1109/ACCESS.2017.2750743.

[19] H. W. Dong, C. Mills, R. T. Knight, J. W. Y. Kam. Detection of mind wandering using EEG: Within and across individuals. *PLoS One*, vol. 16, no. 5, Article number e0251490, 2021. DOI: 10.1371/journal.pone.0251490.

[20] Ç. İ. Acı, M. Kaya, Y. Mishchenko. Distinguishing mental attention states of humans via an EEG-based passive BCI using machine learning methods. *Expert Systems with Applications*, vol. 134, pp. 153–166, 2019. DOI: 10.1016/j.eswa.2019.05.057.

[21] B. H. Cho, J. M. Lee, J. H. Ku, D. P. Jang, J. S. Kim, I. Y. Kim, J. H. Lee, S. I. Kim. Attention enhancement system using virtual reality and EEG biofeedback. In *Proceedings of IEEE Virtual Reality 2002*, IEEE, Orlando, USA, pp. 156–163, 2022. DOI: 10.1109/VR.2002.996518.

[22] H. T. Tor, C. P. Ooi, N. S. Lim-Ashworth, J. K. E. Wei, V. Jahmunah, S. L. Oh, U. R. Acharya, D. S. S. Fung. Automated detection of conduct disorder and attention deficit hyperactivity disorder using decomposition and nonlinear techniques with EEG signals. *Computer Methods and Programs in Biomedicine*, vol. 200, Article number 105941, 2021. DOI: 10.1016/j.cmpb.2021.105941.

[23] S. Geirnaert, S. Vandecappelle, E. Alickovic, A. de Cheveigne, E. Lalor, B. T. Meyer, S. Miran, T. Francart, A. Bertrand. Electroencephalography-based auditory attention decoding: Toward neurosteered hearing devices. *IEEE Signal Processing Magazine*, vol. 38, no. 4, pp. 89–102, 2021. DOI: 10.1109/MSP.2021.3075932.

[24] W. L. Zheng, K. P. Gao, G. Li, W. Liu, C. Liu, J. Q. Liu, G. X. Wang, B. L. Lu. Vigilance estimation using a wearable EOG device in real driving environment. *IEEE Transactions on Intelligent Transportation Systems*, vol. 21, no. 1, pp. 170–184, 2020. DOI: 10.1109/TITS.2018.2889962.

[25] F. Marini, C. Lee, J. Wagner, S. Makeig, M. Gola. A comparative evaluation of signal quality between a research-grade and a wireless dry-electrode mobile EEG system. *Journal of Neural Engineering*, vol. 16, no. 5, Article number 054001, 2019. DOI: 10.1088/1741-2552/ab21f2.

[26] W. L. Zheng, B. L. Lu. A multimodal approach to estimating vigilance using EEG and forehead EOG. *Journal of Neural Engineering*, vol. 14, no. 2, Article number 026017, 2017. DOI: 10.1088/1741-2552/aa5a98.

[27] N. Das, T. Francart, A. Bertrand. Auditory attention detection dataset KULeuven (1.1.0), [Online], Available: https://doi.org/10.5281/zenodo.3997352, 2020.

[28] S. A. Fuglsang, D. D. E. Wong, J. Hjortkjær. EEG and audio dataset for auditory attention decoding, [Online], Available: https://doi.org/10.5281/zenodo.1199011, 2018.

[29] C. Reichert, I. F. Tellez Ceja, C. M. Sweeney-Reed, H. J. Heinze, H. Hinrichs, S. Dürschmid. Impact of stimulus features on the performance of a gaze-independent brain-computer interface based on covert spatial attention shifts. *Frontiers in Neuroscience*, vol. 14, Article number 591777, 2020. DOI: 10.3389/fnins.2020.591777.

[30] Y. S. Zhang, H. Cai, L. Nie, P. Xu, S. R. Zhao, C. T. Guan. An end-to-end 3D convolutional neural network for decoding attentive mental state. *Neural Networks*, vol. 144, pp. 129–137, 2021. DOI: 10.1016/j.neunet.2021.08.019.

[31] K. Dhindsa, A. Acai, N. Wagner, D. Bosynak, S. Kelly, M. Bhandari, B. Petrisor, R. R. Sonnadara. Individualized pattern recognition for detecting mind wandering from EEG during live lectures. *PLoS One*, vol. 14, no. 9, Article number e0222276, 2019. DOI: 10.1371/journal.pone.0222276.

[32] M. Jaeger, B. Mirkovic, M. G. Bleichner, S. Debener. Decoding the attended speaker from EEG using adaptive evaluation intervals captures fluctuations in attentional listening. *Frontiers in Neuroscience*, vol. 14, Article number 603, 2020. DOI: 10.3389/fnins.2020.00603.

[33] M. Geravanchizadeh, S. B. Gavgani. Selective auditory attention detection based on effective connectivity by single-trial EEG. *Journal of Neural Engineering*, vol. 17, no. 2, Article number 026021, 2020. DOI: 10.1088/1741-2552/ab7c8d.

[34] G. Ciccarelli, M. Nolan, J. Perricone, P. T. Calamia, S. Haro, J. O'Sullivan, N. Mesgarani, T. F. Quatieri, C. J. Smalt. Comparison of two-talker attention decoding from EEG with nonlinear neural networks and linear methods. *Scientific Reports*, vol. 9, no. 1, Article number 11538, 2019. DOI: 10.1038/s41598-019-47795-0.

[35] N. Das, A. Bertrand, T. Francart. EEG-based auditory attention detection: Boundary conditions for background noise and speaker positions. *Journal of Neural Engineering*, vol. 15, no. 6, Article number 066017, 2018. DOI: 10.1088/1741-2552/aae0a6.

[36] S. A. Fuglsang, T. Dau, J. Hjortkjær. Noise-robust cortical tracking of attended speech in real-world acoustic scenes. *Neuroimage*, vol. 156, pp. 435–444, 2017. DOI: 10.1016/j.neuroimage.2017.04.026.

[37] C. Obuchi, T. Kawase, K. Kaga, Y. Noguchi. Auditory attention ability under dichotic dual-task situation in adults with listening difficulties. *Audiology and Neurotology*, vol. 28, no. 3, pp. 175–182, 2023. DOI: 10.1159/000528050.

[38] P. V. Corkum, L. S. Siegel. Is the continuous performance task a valuable research tool for use with children with attention-deficit-hyperactivity disorder? *Journal of Child Psychology and Psychiatry*, vol. 34, no. 7, pp. 1217–1239, 1993. DOI: 10.1111/j.1469-7610.1993.tb01784.x.

[39] H. E. Rosvold, A. F. Mirsky, I. Sarason, E. D. Bransome Jr, L. H. Beck. A continuous performance test of brain damage. *Journal of Consulting Psychology*, vol. 20, no. 5, pp. 343–350, 1956. DOI: 10.1037/h0043220.

[40] W. S. Helton, P. N. Russell. Feature absence–presence and two theories of lapses of sustained attention. *Psychological Research*, vol. 75, no. 5, pp. 384–392, 2011. DOI: 10.1007/s00426-010-0316-1.

[41] J. Shin, A. von Lühmann, D. W. Kim, J. Mehnert, H. J. Hwang, K. R. Müller. Simultaneous acquisition of EEG and NIRS during cognitive tasks for an open access dataset. *Scientific Data*, vol. 5, Article number 180003, 2018. DOI: 10.1038/sdata.2018.3.

[42] Y. Choi, M. Kim, C. Chun. Effect of temperature on attention ability based on electroencephalogram measurements. *Building and Environment*, vol. 147, pp. 299–304, 2019. DOI: 10.1016/j.buildenv.2018.10.020.

[43] H. L. Wang, X. Wu, L. Yao. Identifying cortical brain dir-

ected connectivity networks from high-density EEG for emotion recognition. *IEEE Transactions on Affective Computing*, vol. 13, no. 3, pp. 1489–1500, 2022. DOI: 10.1109/TAFFC.2020.3006847.

[44] M. G. Bleichner, M. Lundbeck, M. Selisky, F. Minow, M. Jäger, R. Emkes, S. Debener, M. De Vos. Exploring miniaturized EEG electrodes for brain-computer interfaces. An EEG you do not see? *Physiological reports*, vol. 3, no. 4, Article number e12362, 2015. DOI: 10.14814/phy2.12362.

[45] A. Serrano-Barroso, R. Siugzdaite, J. Guerrero-Cubero, A. J. Molina-Cantero, I. M. Gomez-Gonzalez, J. C. Lopez, J. P. Vargas. Detecting attention levels in ADHD children with a video game and the measurement of brain activity with a single-channel BCI headset. *Sensors*, vol. 21, no. 9, Article number 3221, 2021. DOI: 10.3390/s21093221.

[46] J. X. Wang, W. Q. Wang, Z. G. Hou. Toward improving engagement in neural rehabilitation: Attention enhancement based on brain–computer interface and audiovisual feedback. *IEEE Transactions on Cognitive and Developmental Systems*, vol. 12, no. 4, pp. 787–796, 2020. DOI: 10.1109/TCDS.2019.2959055.

[47] S. Q. Cai, E. Z. Su, L. H. Xie, H. Z. Li. EEG-based auditory attention detection via frequency and channel neural attention. *IEEE Transactions on Human-Machine Systems*, vol. 52, no. 2, pp. 256–266, 2022. DOI: 10.1109/THMS.2021.3125283.

[48] S. Aliakbaryhosseinabadi, E. N. Kamavuako, N. Jiang, D. Farina, N. Mrachacz-Kersting. Classification of EEG signals to identify variations in attention during motor task execution. *Journal of Neuroscience Methods*, vol. 284, pp. 27–34, 2017. DOI: 10.1016/j.jneumeth.2017.04.008.

[49] F. G. Luna, R. Román-Caballero, P. Barttfeld, J. Lupiáñez, E. Martín-Arévalo. A high-definition tDCS and EEG study on attention and vigilance: Brain stimulation mitigates the executive but not the arousal vigilance decrement. *Neuropsychologia*, vol. 142, Article number 107447, 2020. DOI: 10.1016/j.neuropsychologia.2020.107447.

[50] I. Stancin, M. Cifrek, A. Jovic. A review of EEG signal features and their application in driver drowsiness detection systems. *Sensors*, vol. 21, no. 11, Article number 3786, 2021. DOI: 10.3390/s21113786.

[51] A. Bednar, E. C. Lalor. Where is the cocktail party? Decoding locations of attended and unattended moving sound sources using EEG. *NeuroImage*, vol. 205, Article number 116283, 2020. DOI: 10.1016/j.neuroimage.2019.116283.

[52] L. Fiedler, M. Wöstmann, C. Graversen, A. Brandmeyer, T. Lunner, J. Obleser. Single-channel in-ear-EEG detects the focus of auditory attention to concurrent tone streams and mixed speech. *Journal of Neural Engineering*, vol. 14, no. 3, Article number 036020, 2017. DOI: 10.1088/1741-2552/aa66dd.

[53] D. Ni, S. Wang, G. C. Liu. The EEG-based attention analysis in multimedia m-learning. *Computational and Mathematical Methods in Medicine*, vol. 2020, Article number 4837291, 2020. DOI: 10.1155/2020/4837291.

[54] M. Torkamani-Azar, S. D. Kanik, S. Aydin, M. Cetin. Prediction of reaction time and vigilance variability from spatio-spectral features of resting-state EEG in a long sustained attention task. *IEEE Journal of Biomedical and Health Informatics*, vol. 24, no. 9, pp. 2550–2558, 2020. DOI: 10.1109/JBHI.2020.2980056.

[55] L. W. Ko, O. Komarov, W. D. Hairston, T. P. Jung, C. T. Lin. Sustained attention in real classroom settings: An EEG study. *Frontiers in Human Neuroscience*, vol. 11, Article number 388, 2017. DOI: 10.3389/fnhum.2017.00388.

[56] M. Chen, Y. Cao, R. Wang, Y. Li, D. Wu, Z. C. Liu. DeepFocus: Deep encoding brainwaves and emotions with multi-scenario behavior analytics for human attention enhancement. *IEEE Network*, vol. 33, no. 6, pp. 70–77, 2019. DOI: 10.1109/MNET.001.1900054.

[57] A. Farahat, C. Reichert, C. M. Sweeney-Reed, H. Hinrichs. Convolutional neural networks for decoding of covert attention focus and saliency maps for EEG feature visualization. *Journal of Neural Engineering*, vol. 16, no. 6, Article number 066010, 2019. DOI: 10.1088/1741-2552/ab3bb4.

[58] S. Geirnaert, T. Francart, A. Bertrand. Fast EEG-based decoding of the directional focus of auditory attention using common spatial patterns. *IEEE Transactions on Biomedical Engineering*, vol. 68, no. 5, pp. 1557–1568, 2021. DOI: 10.1109/TBME.2020.3033446.

[59] L. M. Vortmann, F. Kroll, F. Putze. EEG-based classification of internally-and externally-directed attention in an augmented reality paradigm. *Frontiers in Human Neuroscience*, vol. 13, Article number 348, 2019. DOI: 10.3389/fnhum.2019.00348.

[60] D. P. Kingma, J. Ba. Adam: A method for stochastic optimization. In *Proceedings of the 3rd International Conference on Learning Representations*, San Diego, USA, 2014. DOI: 10.48550/arXiv.1412.6980.

[61] J. Vanthornhout, L. Decruy, T. Francart. Effect of task and attention on neural tracking of speech. *Frontiers in Neuroscience*, vol. 13, Article number 977, 2019. DOI: 10.3389/fnins.2019.00977.

[62] C. Y. Jin, J. P. Borst, M. K. van Vugt. Predicting task-general mind-wandering with EEG. *Cognitive, Affective, & Behavioral Neuroscience*, vol. 19, no. 4, pp. 1059–1073, 2019. DOI: 10.3758/s13415-019-00707-1.

[63] D. Hölle, J. Meekes, M. G. Bleichner. Mobile ear-EEG to study auditory attention in everyday life: Auditory attention in everyday life. *Behavior Research Methods*, vol. 53, no. 5, pp. 2025–2036, 2021. DOI: 10.3758/s13428-021-01538-0.

[64] D. van Son, F. M. De Blasio, J. S. Fogarty, A. Angelidis, R. J. Barry, P. Putman. Frontal EEG theta/beta ratio during mind wandering episodes. *Biological Psychology*, vol. 140, pp. 19–27, 2019. DOI: 10.1016/j.biopsycho.2018.11.003.

[65] C. J. Peng, Y. C. Chen, C. C. Chen, S. J. Chen, B. Cagneau, L. Chassagne. An EEG-based attentiveness recognition system using Hilbert–Huang transform and support vector machine. *Journal of Medical and Biological Engineering*, vol. 40, no. 2, pp. 230–238, 2020. DOI: 10.1007/s40846-019-00500-y.

[66] K. K. Ang, Z. Y. Chin, H. H. Zhang, C. T. Guan. Filter bank common spatial pattern (FBCSP) in brain-computer interface. In *Proceedings of IEEE International Joint Conference on Neural Networks (IEEE World Congress on Computational Intelligence)*, IEEE, Hong Kong, China, pp. 2390–2397, 2008. DOI: 10.1109/IJCNN.2008.

4634130.

[67] I. Winkler, S. Haufe, M. Tangermann. Automatic classification of artifactual ICA-components for artifact removal in EEG signals. *Behavioral and Brain Functions*, vol. 7, Article number 30, 2011. DOI: 10.1186/1744-9081-7-30.

[68] A. Subasi, M. I. Gursoy. EEG signal classification using PCA, ICA, LDA and support vector machines. *Expert Systems with Applications*, vol. 37, no. 12, pp. 8659–8666, 2010. DOI: 10.1016/j.eswa.2010.06.065.

[69] C. Y. Chang, S. H. Hsu, L. Pion-Tonachini, T. P. Jung. Evaluation of artifact subspace reconstruction for automatic artifact components removal in multi-channel EEG recordings. *IEEE Transactions on Biomedical Engineering*, vol. 67, no. 4, pp. 1114–1121, 2020. DOI: 10.1109/TBME.2019.2930186.

[70] A. S. Al-Fahoum, A. A. Al-Fraihat. Methods of EEG signal features extraction using linear analysis in frequency and time-frequency domains. *International Scholarly Research Notices*, vol. 2014, Article number 730218, 2014. DOI: 10.1155/2014/730218.

[71] E. T. M. Beltrán, M. Q. Pérez, S. L. Bernal, G. M. Pérez, A. H. Celdrán. SAFECAR: A brain–computer interface and intelligent framework to detect drivers′ distractions. *Expert Systems with Applications*, vol. 203, Article number 117402, 2022. DOI: 10.1016/j.eswa.2022.117402.

[72] G. F. Li, W. Q. Yan, S. Li, X. D. Qu, W. B. Chu, D. P. Cao. A temporal–spatial deep learning approach for driver distraction detection based on EEG signals. *IEEE Transactions on Automation Science and Engineering*, vol. 19, no. 4, pp. 2665–2677, 2022. DOI: 10.1109/TASE.2021.3088897.

[73] N. Das, J. Zegers, H. van Hamme, T. Francart, A. Bertrand. Linear versus deep learning methods for noisy speech separation for EEG-informed attention decoding. *Journal of Neural Engineering*, vol. 17, no. 4, Article number 046039, 2020. DOI: 10.1088/1741-2552/aba6f8.

[74] C. J. Fan, Y. Peng, S. L. Peng, H. H. Zhang, Y. K. Wu, S. Kwong. Detection of train driver fatigue and distraction based on forehead EEG: A time-series ensemble learning method. *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 8, pp. 13559–13569, 2022. DOI: 10.1109/TITS.2021.3125737.

[75] S. Vandecappelle, L. Deckers, N. Das, A. H. Ansari, A. Bertrand, T. Francart. EEG-based detection of the locus of auditory attention with convolutional neural networks. *eLife*, vol. 10, Article number e56481, 2021. DOI: 10.7554/eLife.56481.

[76] W. Nogueira, H. Dolhopiatenko, I. Schierholz, A. Büchner, B. Mirkovic, M. G. Bleichner, S. Debener. Decoding selective attention in normal hearing listeners and bilateral cochlear implant users with concealed ear EEG. *Frontiers in Neuroscience*, vol. 13, Article number 720, 2019. DOI: 10.3389/fnins.2019.00720.

[77] A. Apicella, P. Arpaia, M. Frosolone, N. Moccaldi. High-wearable EEG-based distraction detection in motor rehabilitation. *Scientific Reports*, vol. 11, no. 1, Article number 5297, 2021. DOI: 10.1038/s41598-021-84447-8.

[78] E. S. Teoh, E. C. Lalor. EEG decoding of the target speaker in a cocktail party scenario: Considerations regarding dynamic switching of talker location. *Journal of Neural Engineering*, vol. 16, no. 3, Article number 036017, 2019. DOI: 10.1088/1741-2552/ab0cf1.

[79] W. Nogueira, G. Cosatti, I. Schierholz, M. Egger, B. Mirkovic, A. Büchner. Toward decoding selective attention from single-trial EEG data in cochlear implant users. *IEEE Transactions on Biomedical Engineering*, vol. 67, no. 1, pp. 38–49, 2020. DOI: 10.1109/TBME.2019.2907638.

[80] D. Wang, J. Y. Chen, D. Zhao, F. Dai, C. J. Zheng, X. B. Wu. Monitoring workers' attention and vigilance in construction activities through a wireless and wearable electroencephalography system. *Automation in Construction*, vol. 82, pp. 122–137, 2017. DOI: 10.1016/j.autcon.2017.02.001.

[81] I. García-Cordero, S. Esteves, E. P. Mikulan, E. Hesse, F. H. Baglivo, W. Silva, M. del Carmen García, E. Vaucheret, C. Ciraolo, H. S. García, F. Adolfi, M. Pietto, E. Herrera, A. Legaz, F. Manes, A. M. García, M. Sigman, T. A. Bekinschtein, A. Ibáñez, L. Sedeño. Attention, in and out: Scalp-level and intracranial EEG correlates of interoception and exteroception. *Frontiers in Neuroscience*, vol. 11, Article number 411, 2017. DOI: 10.3389/fnins.2017.00411.

[82] D. D. E. Wong, S. A. Fuglsang, J. Hjortkjær, E. Ceolini, M. Slaney, A. De Cheveigné. A comparison of regularization methods in forward and backward models for auditory attention decoding. *Frontiers in Neuroscience*, vol. 12, Article number 531, 2018. DOI: 10.3389/fnins.2018.00531.

[83] H. S. Chiang, K. L. Hsiao, L. C. Liu. EEG-based detection model for evaluating and improving learning attention. *Journal of Medical and Biological Engineering*, vol. 38, no. 6, pp. 847–856, 2018. DOI: 10.1007/s40846-017-0344-z.

[84] Y. Y. Zhou, S. Huang, Z. M. Xu, P. P. Wang, X. Wu, D. Q. Zhang. Cognitive workload recognition using EEG signals and machine learning: A review. *IEEE Transactions on Cognitive and Developmental Systems*, vol. 14, no. 3, pp. 799–818, 2022. DOI: 10.1109/TCDS.2021.3090217.

[85] M. Saeidi, W. Karwowski, F. V. Farahani, K. Fiok, R. Taiar, P. A. Hancock, A. Al-Juaid. Neural decoding of EEG signals with machine learning: A systematic review. *Brain Sciences*, vol. 11, no. 11, Article number 1525, 2021. DOI: 10.3390/brainsci11111525.

[86] S. Miran, S. Akram, A. Sheikhattar, J. Z. Simon, T. Zhang, B. Babadi. Real-time tracking of selective auditory attention from M/EEG: A bayesian filtering approach. *Frontiers in Neuroscience*, vol. 12, Article number 262, 2018. DOI: 10.3389/fnins.2018.00262.

[87] M. K. Hazrati, A. Erfanian. An online EEG-based brain–computer interface for controlling hand grasp using an adaptive probabilistic neural network. *Medical Engineering & Physics*, vol. 32, no. 7, pp. 730–739, 2010. DOI: 10.1016/j.medengphy.2010.04.016.

[88] H. R. Wang, Y. R. Sun, F. Wang, L. Cao, W. Zhou, Z. J. Wang, S. Y. Chen. Cross-subject assistance: Inter-and intra-subject maximal correlation for enhancing the performance of SSVEP-based BCIs. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 29, pp. 517–526, 2021. DOI: 10.1109/TNSRE.2021.3057938.

[89] R. L. Li, L. P. Wang, O. Sourina. Subject matching for cross-subject EEG-based recognition of driver states related to situation awareness. *Methods*, vol. 202, pp. 136–143, 2022. DOI: 10.1016/j.ymeth.2021.04.009.

[90] X. Y. Jiang, J. X. Huang, S. Jin, S. J. Lu. Domain generalization via balancing training difficulty and model capabil-

ity. In *Proceedings of IEEE/CVF International Conference on Computer Vision*, IEEE, Paris, France, pp. 18947–18957, 2023. DOI: 10.1109/ICCV51070.2023.01741.

Qianru Sun received the B.Sc. degree in computer science and technology from Nanjing University of Aeronautics and Astronautics (NUAA), China in 2022. She is currently a master student in computer science and technology at NUAA, China.

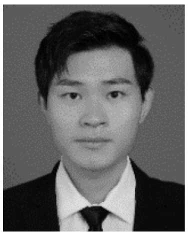
Her research interests include EEG analysis, attention detection, and machine learning.

E-mail: sqq666@nuaa.edu.cn
ORCID iD: 0009-0000-3924-9725

Yueying Zhou received the Ph.D. degree in computer science from Nanjing University of Aeronautics and Astronautics (NUAA), China in 2023. She joined the School of Mathematics Science of Liaocheng University, China, as a lecturer in 2023.

Her research interests include brain-computer interface, EEG analysis, and machine learning.

E-mail: zhouyueying@nuaa.edu.cn



Peiliang Gong received the M.Sc. degree from Beijing Jiaotong University, China in 2021. He is currently a Ph.D. degree candidate with College of Computer Science and Technology, Nanjing University of Aeronautics and Astronautics, China.

His research interests include machine learning, brain computer interface, and EEG analysis.

E-mail: plgong@nuaa.edu.cn

Daoqiang Zhang (Member, IEEE) received the B.Sc. and Ph.D. degrees in computer science from Nanjing University of Aeronautics and Astronautics (NUAA), China in 1999 and 2004, respectively. He joined the Department of Computer Science and Engineering, NUAA as a lecturer in 2004, where he is currently a professor. He is a Fellow of the International Association for Pattern Recognition (IAPR).

His research interests include machine learning, pattern recognition, data mining, and medical image analysis. In these areas, he has published more than 200 scientific articles with more than 15 000 citations by Google Scholar.

E-mail: dqzhang@nuaa.edu.cn (Corresponding author)